Robust and Reproducible Prediction of HDL Cholesterol Under Outcome Perturbation: An NHANES Machine Learning Study

Francis Okyere | Sherif Mohammed
Department of Statistics, Florida State University

Abstract

This study evaluates the robustness of predictive models for HDL cholesterol using NHANES data under systematically injected Gaussian outcome perturbation. Linear, regularized, and ensemble methods were compared using cross-validation. XGBoost achieved the strongest predictive accuracy and robustness across increasing noise levels, while anthropometric and demographic predictors remained stable, supporting reliable, interpretable, and reproducible health prediction.

1. Introduction

High-density lipoprotein cholesterol (HDL-C) is a key biomarker for cardiovascular health and metabolic risk (Gordon et al., 1989). Accurate prediction of HDL-C using large-scale observational health data supports risk stratification and personalized interventions (Hastie, Tibshirani, & Friedman, 2009).

In practice, outcomes may be corrupted by random noise arising from laboratory variability, reporting error, or privacy-preserving perturbations (Dwork & Roth, 2014). Such noise can degrade predictive accuracy and destabilize model selection. This study investigates the robustness of multiple statistical and machine learning models for predicting HDL-C using NHANES data under increasing levels of outcome perturbation. Our analysis emphasizes model stability, interpretability, and reproducibility beyond leaderboard optimization.

2. Data and Preprocessing

The training data were derived from the National Health and Nutrition Examination Survey (NHANES), which provides nationally representative health and nutrition data for the U.S. population (National Center for Health Statistics [NCHS], 2024). The outcome variable was `LBDHDD_outcome` (mg/dL). Predictors included dietary intake, demographics, and anthropometric measures.

Missing values were imputed using median imputation for continuous variables and mode imputation for categorical variables. Categorical variables were encoded using one-hot encoding, and numeric predictors were standardized to support stable estimation and fair regularization.

To strengthen model parsimony and reduce redundancy, we implemented systematic pre-model feature screening. Near-zero variance predictors were removed, linear combinations were eliminated to address perfect multicollinearity, and highly correlated predictors were filtered using a predefined correlation threshold. All preprocessing steps were performed within a unified pipeline fit exclusively on the training data to prevent data leakage.

Additional Gaussian noise was injected to assess sensitivity:

$$Y^{(\sigma)} = Y + \epsilon, \qquad \epsilon \sim N(0, \sigma^2),$$

with $\sigma \in \{0, 0.5, 1, 2, 3, 5\}$.

3. Methodology: Modeling, Validation, and Stability

3.1 Candidate Models

We evaluated:

- Ordinary Least Squares (OLS)
- Ridge Regression (Hoerl & Kennard, 1970)
- Lasso Regression (Tibshirani, 1996)
- Elastic Net (Zou & Hastie, 2005)
- Random Forest (Breiman, 2001)
- XGBoost (Chen & Guestrin, 2016)

3.2 Validation and Tuning Strategy

A fixed K-fold cross-validation scheme was used to obtain out-of-fold (OOF) predictions for unbiased performance estimation. Hyperparameters were tuned at baseline noise ($\sigma = 0$) and subsequently held fixed across all noise levels to isolate the impact of outcome perturbation (Hastie et al., 2009).

Performance was evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2, with RMSE serving as the primary competition ranking metric. To assess dimensionality effects, we additionally compared predictive performance using a reduced feature set selected via Lasso at λ_{1se} against the screened full feature set.

3.3 Feature Stability

Elastic Net stability was assessed using bootstrap resampling to quantify how consistently predictors were selected across repeated perturbations. Selection frequency and coefficient magnitude were aggregated into a stability score.

4. Results and Robustness Analysis

The HDL-C outcome exhibits an approximately unimodal distribution centered near 55 mg/dL, with moderate dispersion and limited skewness. At baseline noise ($\sigma = 0$), XGBoost achieved the lowest RMSE and MAE and the highest R^2, followed by Random Forest and regularized linear models. Ordinary Least Squares showed inferior performance, reflecting sensitivity to multicollinearity and high-dimensional predictor structure. As outcome noise increased, all models exhibited monotonic degradation in predictive performance. OLS was most sensitive to perturbation, whereas XGBoost and Random Forest degraded more gradually, demonstrating superior robustness. Regularized linear models displayed intermediate behavior. Performance using the reduced Lasso-selected feature set yielded comparable RMSE to the full screened feature set, indicating that ensemble robustness is driven by stable signal extraction rather than redundant predictors. Feature stability analysis revealed that waist circumference, gender, alcohol intake, age, and income-to-poverty ratio were consistently selected across noise levels (Figure 1).

5. Final Model and Prediction Generation

Based on cross-validation and robustness analysis, XGBoost was selected as the final model. The model was trained on the full training dataset and used to generate predictions for the unlabeled test set. The submission file `pred.csv` contains exactly one column named `pred`, aligned with the test dataset order.

Figure 1: Combined visualization of outcome distribution, baseline cross-validated performance, performance degradation under increasing Gaussian noise, and robust feature importance.

5.1 Code Availability and Reproducibility

All preprocessing steps—including systematic feature screening, model tuning, cross-validation, and prediction generation—were implemented within a fully automated and documented pipeline. The complete source code and instructions for reproducing all analyses are publicly available at:

https://github.com/KojoOkyere/hdl-noise-robust-prediction

Running `main_pipeline.R` from the project root directory reproduces all reported tables, figures, and predictions.

6. Discussion and Conclusion

Ensemble-based models substantially outperformed classical linear models, reflecting their ability to capture nonlinear interactions and complex predictor dependencies (Breiman, 2001; Chen & Guestrin, 2016). Regularized regression demonstrated improved robustness relative to OLS, consistent with bias–variance trade-off theory (Hastie et al., 2009).

The strong stability of anthropometric and demographic predictors aligns with established relationships between adiposity, age, lifestyle, and lipid metabolism. These findings support the deployment of ensemble and shrinkage methods for predictive modeling in noisy or privacy-preserving health environments.

Key findings:

- Best baseline model: XGBoost
- Most robust under noise: XGBoost (RF competitive)
- Stable predictors: waist circumference, gender, alcohol intake, age, income

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